

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 40%

1. Real-Time System Design Trade-off

A surveillance system must operate under constraints:

- Viola-Jones: fast but less accurate (78%)
- YOLO: moderate speed, good accuracy (92%)
- Deep Learning: high accuracy (95%) but slow

Compare the techniques and evaluate which should be selected under strict latency constraints while maintaining acceptable accuracy. Justify your decision considering deployment conditions.

REAL-TIME SYSTEM DESIGN TRADE-OFF

A surveillance system must balance accuracy, processing speed, robustness, and computational cost. The three techniques can be compared as follows:

Technique	Accuracy	Speed	Computational Requirement	Suitability
Viola-Jones	78%	Very Fast	Low	Real-time, low-resource systems
YOLO	92%	Moderate/Fast	Medium-High	Real-time surveillance
Deep Learning	95%	Slow	High	High-accuracy, non-latency-critical systems

Evaluation:

- **Viola-Jones** has the lowest computational cost and very low latency. However, its **78% accuracy** may result in more false detections or missed objects, reducing reliability in complex surveillance environments.
- **Deep Learning** provides the highest accuracy (**95%**) and better recognition capability under challenging conditions. However, its slower processing and high computational requirements make it unsuitable when strict latency is required.
- **YOLO** provides a strong compromise, achieving **92% accuracy** with moderate processing speed. It can detect objects efficiently while maintaining relatively high recognition accuracy.

Decision:

Under strict latency constraints, YOLO should be selected. Although Viola-Jones is faster, its 78% accuracy is considerably lower. YOLO provides 14 percentage points higher accuracy than Viola-Jones while still supporting real-time or near-real-time processing.

Deployment Justification:

For a surveillance system deployed on a GPU-enabled edge device or moderately powerful computing system, YOLO is the most appropriate choice because it offers a good accuracy–latency trade-off. If the hardware is extremely resource-constrained and latency is the absolute priority, Viola-Jones could be considered. Deep Learning would be preferred only when accuracy is more important than response time.

Conclusion:

YOLO is the best overall choice because it maintains high accuracy (**92%**) while providing substantially better speed than a high-accuracy deep-learning approach. It offers the most practical balance for real-time surveillance applications.

5. Motion Estimation vs Computational Cost

Two methods:

- Method A: low error (5%), high computation
- Method B: moderate error (10%), low computation

Compare the methods and evaluate their suitability for real-time vs offline applications.

Justify your decision.

MOTION ESTIMATION VS COMPUTATIONAL COST

Motion estimation methods can be compared based on **estimation error, computational cost, processing speed, and application requirements.**

Factor	Method A	Method B
Motion estimation error	5% (Low)	10% (Moderate)
Computational cost	High	Low
Processing speed	Slow	Fast
Accuracy	High	Moderate
Real-time suitability	Moderate/Low	High
Offline suitability	High	Moderate

Evaluation:

- **Method A** provides a lower error of **5%**, meaning it estimates object motion more accurately. However, its high computational requirement increases processing time.
- **Method B** has a higher error of **10%**, but its low computational cost allows faster processing and makes it more efficient for systems with limited resources.

For real-time applications:

Method B is more suitable because real-time systems prioritize low latency and fast processing. Applications such as live surveillance, autonomous systems, and video conferencing need motion information immediately. A 10% error may be acceptable if it allows the system to respond quickly.

For offline applications:

Method A is more suitable because offline processing does not have strict latency requirements. The system can spend more computational resources to achieve the lower 5% error. This is useful in applications such as video analysis, scientific research, motion tracking, and post-event surveillance analysis.

Conclusion:

Method B should be selected for real-time applications due to its low computational cost and faster processing, while **Method A** should be selected for offline applications where higher accuracy is more important than processing speed. Thus, the choice depends on the trade-off between **accuracy and computational efficiency**.

9. YOLO Performance across Scales

Detection results:

- Large objects: 95%
- Medium objects: 85%
- Small objects: 60%

Compare performance across scales and evaluate whether YOLO alone is sufficient for multi-scale detection tasks. Justify improvements.

YOLO PERFORMANCE ACROSS SCALES

YOLO shows different detection performance depending on the **object scale**. The results can be compared as follows:

Object Scale	Detection Accuracy	Evaluation
Large objects	95%	Excellent
Medium objects	85%	Good
Small objects	60%	Poor

Comparison and Evaluation:

- **Large objects:** YOLO achieves **95% accuracy**, indicating that it can reliably detect objects with sufficient visual features and larger spatial representation.
- **Medium objects:** Accuracy decreases to **85%**, showing that detection becomes more challenging as the object occupies fewer pixels.
- **Small objects:** Accuracy drops significantly to **60%**. Small objects contain limited visual information and may lose important features during feature extraction and downsampling.

The performance difference between large and small objects is **35 percentage points**, which indicates a significant scale-related limitation.

Is YOLO Alone Sufficient?

YOLO alone may **not be sufficient for applications requiring reliable multi-scale detection**, especially when small objects are important. Although it performs very well on large objects

and reasonably well on medium objects, the **60% accuracy for small objects** can lead to missed detections and false negatives.

Possible Improvements:

1. **Multi-scale feature extraction:** Use Feature Pyramid Networks (FPN) or Path Aggregation Networks (PAN) to combine features from different resolutions.
2. **Higher input resolution:** Increasing the input image resolution can preserve more information about small objects.
3. **Small-object detection heads:** Add specialized detection layers operating at higher-resolution feature maps.
4. **Data augmentation:** Techniques such as random cropping, scaling, and mosaic augmentation can improve the model's ability to recognize objects at different scales.
5. **Anchor/label optimization:** Properly tuning anchor sizes or detection parameters for small objects can improve localization.

Conclusion:

YOLO provides excellent large-object detection (95%) and good medium-object detection (85%), but its 60% small-object accuracy indicates that additional multi-scale techniques are required. Therefore, for critical multi-scale applications such as surveillance, autonomous driving, and aerial imagery, an enhanced YOLO architecture with multi-scale feature fusion and higher-resolution detection would be a better choice than using basic YOLO alone.

19. Motion Tracking Accuracy Comparison

Results:

Method	Accuracy	Stability
A	High	Low
B	Moderate	High

Compare and evaluate which is suitable for long-term tracking.

MOTION TRACKING ACCURACY COMPARISON

The two motion-tracking methods can be compared based on **accuracy and stability**:

Method	Accuracy	Stability	Overall Evaluation
Method A	High	Low	Accurate but inconsistent
Method B	Moderate	High	Consistent and reliable

Comparison:

- **Method A** provides **high tracking accuracy**, meaning it can locate the target precisely. However, its **low stability** may cause tracking failures, jitter, or loss of the target over extended periods.
- **Method B** has **moderate accuracy**, but its **high stability** allows it to maintain the target's identity and position consistently over time.

Evaluation for Long-Term Tracking:

For **long-term tracking**, **Method B is more suitable**. Long-term tracking requires consistent performance even when there are changes in object position, lighting, partial occlusion, or background conditions. High stability reduces the possibility of frequent tracking loss. Although Method A has better instantaneous accuracy, its low stability makes it less reliable for continuous tracking.

Conclusion:

Method B should be selected for long-term motion tracking because stability is more important than maximum instantaneous accuracy. A stable tracker can maintain continuous object tracking over long periods, whereas a highly accurate but unstable method may frequently lose the target.

20. Surveillance Model Selection

Results:

Model	Accuracy	Latency
A	82%	30 ms
B	90%	70 ms
C	95%	120 ms

Compare and evaluate under strict latency constraints.

SURVEILLANCE MODEL SELECTION

The models can be compared based on **accuracy and latency**:

Model	Accuracy	Latency	Evaluation
A	82%	30 ms	Fastest, lower accuracy
B	90%	70 ms	Good balance
C	95%	120 ms	Highest accuracy, slowest

Comparison and Evaluation:

- **Model A** has the lowest latency of **30 ms**, making it highly suitable for fast real-time surveillance. However, its accuracy is only **82%**, which may result in more missed or incorrect detections.
- **Model B** provides **90% accuracy** with **70 ms latency**, offering a better balance between detection performance and response time.
- **Model C** achieves the highest accuracy of **95%**, but its **120 ms latency** makes it unsuitable when the system has strict response-time requirements.

Decision:

Under **strict latency constraints**, **Model A should be selected** because its 30 ms latency allows the surveillance system to respond quickly. In applications such as real-time security monitoring, rapid detection and response may be more important than achieving maximum accuracy.

However, if the latency limit permits around **70 ms**, **Model B would be the preferred practical choice** because it significantly improves accuracy from 82% to 90% while maintaining reasonably low latency.

Conclusion:

For **very strict latency requirements**, **Model A is the best choice** due to its 30 ms response time. If a slightly higher latency is acceptable, **Model B provides the best accuracy–latency**

trade-off. Model C should be selected only when accuracy is more important than real-time response.

21. Noise Impact on Detection

Results:

Noise Level	Accuracy
Low	92%
Medium	80%
High	60%

Compare performance and evaluate system robustness.

NOISE IMPACT ON DETECTION

The detection system's performance decreases as the **noise level increases**.

Noise Level	Accuracy	Performance
Low	92%	Excellent
Medium	80%	Good
High	60%	Poor

Comparison and Evaluation

- Under **low noise**, the system achieves **92% accuracy**, indicating reliable detection when image quality is good.
- Under **medium noise**, accuracy decreases to **80%**, showing that noise begins to interfere with important image features.
- Under **high noise**, accuracy falls significantly to **60%**. This indicates that the system has difficulty distinguishing objects from noisy backgrounds.

The accuracy decreases by **32 percentage points** from low-noise to high-noise conditions. Therefore, the system's robustness to noise is **limited**.

Improvements for Robustness

The system can be made more robust by applying preprocessing techniques such as:

1. **Gaussian or Median Filtering** – Reduces random noise while preserving important image information.
2. **Image Enhancement** – Improves contrast and visibility of relevant features.
3. **Adaptive Thresholding** – Helps separate objects from noisy backgrounds.
4. **Noise-Augmented Training** – Training the detection model with noisy images can improve its ability to handle real-world conditions.

5. **Robust Feature Extraction** – Using features that are less sensitive to noise can improve detection reliability.

Conclusion

The system performs well under low noise (92%) but its accuracy drops considerably under high noise (60%). Therefore, it is not highly robust to noisy conditions in its current form. Applying effective preprocessing and training with noise-augmented data can improve its robustness and make it more suitable for real-world surveillance and computer vision applications.

22. Depth Estimation Error Analysis

Results:

Method	Error
A	0.5 m
B	1.2 m
C	0.8 m

Compare and evaluate accuracy for real-world applications.

DEPTH ESTIMATION ERROR ANALYSIS

The accuracy of the depth estimation methods can be compared based on their estimation error. A lower error indicates higher accuracy.

Method	Depth Estimation Error	Accuracy
A	0.5 m	Highest
B	1.2 m	Lowest
C	0.8 m	Moderate

Comparison and Evaluation

- **Method A** has the lowest error of **0.5 m**, making it the most accurate method for estimating object depth.
- **Method C** has an error of **0.8 m**, providing moderate accuracy and potentially acceptable performance for less safety-critical applications.
- **Method B** has the highest error of **1.2 m**, making it the least accurate and less reliable when precise distance estimation is required.

The error of Method B is **0.7 m higher than Method A**, indicating a significant reduction in depth estimation reliability.

Suitability for Real-World Applications

For applications such as autonomous vehicles, robotics, obstacle avoidance, and 3D scene understanding, Method A is the preferred choice because its lower error provides more precise distance information.

Method C can be considered when moderate accuracy is sufficient and computational or implementation requirements favor it. Method B would be less suitable for applications requiring precise depth estimation, particularly safety-critical systems.

Conclusion

Method A should be selected for real-world applications requiring high depth accuracy, as it has the lowest estimation error of 0.5 m. Method C is a reasonable alternative, while Method B is least suitable due to its relatively high 1.2 m error.

26. Feature Detection under Illumination Changes

Results:

Method	Normal	Low Light
A	90%	60%
B	85%	75%

Compare and evaluate robustness.

FEATURE DETECTION UNDER ILLUMINATION CHANGES

The robustness of the two methods can be evaluated by comparing their performance under **normal and low-light conditions**.

Method	Normal Light	Low Light	Accuracy Drop
A	90%	60%	30%
B	85%	75%	10%

Comparison and Evaluation

- **Method A** achieves a higher accuracy of **90%** under normal lighting. However, its accuracy drops sharply to **60%** in low-light conditions.
- **Method B** achieves **85%** under normal lighting and **75%** under low light. Although its normal-light accuracy is lower than Method A, it experiences only a **10 percentage-point decrease**, demonstrating greater stability.
- Method A has a **30 percentage-point drop**, indicating that it is highly sensitive to illumination changes.
- Method B has only a **10 percentage-point drop**, showing significantly better robustness.

Decision

For applications where lighting conditions vary, Method B is the better choice. Its slightly lower performance under normal lighting is compensated by its much better performance in low-light environments.

Method A would be suitable when lighting can be controlled or remains consistently normal, while Method B is more appropriate for surveillance, outdoor monitoring, and nighttime computer vision applications.

Conclusion

Method B is more robust to illumination changes because its accuracy decreases only from 85% to 75%, whereas Method A decreases from 90% to 60%. Therefore, Method B should be preferred for real-world applications with unpredictable lighting conditions.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: